

Subgroups of prescribed power-of-two order in symmetric groups

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Submitted by tienxion. 4 September 2026. A partial result relevant to Erdős 1163. This note counts actual subgroups, not conjugacy classes, and does not determine the order distribution of a uniformly chosen unrestricted subgroup. Developed with OpenAI GPT-6 assistance, including independent agent proof audits. No claim of novelty or prior expert endorsement is made.

Let

$$a_{n,j} = \#\{H \leq S_n : |H| = 2^j\}.$$

Uniformly for integers $\lfloor n/4 \rfloor \leq j \leq \lfloor n/2 \rfloor$,

$$\log_2 a_{n,j} \geq \frac{n^2}{16} + \left(\frac{7n}{8} - \left\lfloor j - \frac{3n}{8} \right\rfloor \right) \log_2 n - O(n). \quad (1)$$

The groups supplying this bound have nilpotency class at most two, exponent dividing four, and orbits of size at most eight. Combining (1) with the upper bound for **all** 2-subgroups in Roney-Dougal-Tracey, [Theorem 2](#), gives the uniform conclusion

$$\log_2 a_{n,j} = \frac{n^2}{16} + O(n \log n). \quad (2)$$

Thus every prescribed order in this interval attains the full quadratic counting exponent. The remaining error can still change relative probabilities substantially; (2) does not imply equidistribution.

An explicit eight-point group

On $\mathbb{F}_2^2 \times \mathbb{F}_2$, let B_0 consist of the flips

$$(i, e) \mapsto (i, e + b_i), \quad \sum_i b_i = 0.$$

Let $V = \mathbb{F}_2^2$ translate the first coordinate, and put $E = B_0 \rtimes V$. It has order 32 and acts transitively: translations move between fibers, while even flips can interchange the two points of any chosen fiber. A nonzero translation exchanges the four indices in two pairs. For even-weight b , the two pair sums agree, so $b + v(b)$ is zero or the all-one vector $\mathbf{1}$; both values occur. Consequently

$$E' = \langle \mathbf{1} \rangle \cong C_2, \quad E/E' \cong \mathbb{F}_2^4.$$

The square formula $(b, v)^2 = (b + v(b), 0)$ shows that all squares lie in the central derived group, proving the class and exponent assertions. This classical degree-eight factor appears in Kovács-Praeger, [Finite permutation groups with large abelian quotients](#), §2.

Construction at every prescribed order

Assume $n \geq 16$, and write

$$m = \lfloor n/2 \rfloor, \quad f = \lfloor m/2 \rfloor, \quad a = m - 2f \in \{0, 1\}, \quad \delta = n - 2m \in \{0, 1\}.$$

Take $k = f$, except when $m = 2f + 1$ and $j = m$, in which case take $k = f + 1$. Then $k \geq 4$, $k(m - k) = \lfloor m^2/4 \rfloor$, and $z = j - k$ lies between zero and f . Set

$$c = \min(z, f - z), \quad b = f - 2c, \quad d = z - c.$$

These are nonnegative integers satisfying

$$a + 2b + 4c = m, \quad c + d = z, \quad d \in \{0, b\}.$$

Partition the labels into δ singletons, a pairs, b four-point blocks, and c eight-point blocks. Use C_2 on each pair and E on each eight-point block. On every four-point block use the regular Klein four-group if $d = 0$, or the transitive dihedral group of order eight if $d = b$. When $b = 0$, this distinction is immaterial. Transport one fixed model to each labelled block using its increasing ordering.

Their direct product P satisfies

$$P/P' \cong \mathbb{F}_2^m, \quad |P'| = 2^z.$$

Every quotient factor has dimension at most four. Choose a linear surjection from $\mathbb{F}_2^4$ onto each factor. The image of their combined map surjects onto every factor and has dimension at most four; extend it to a four-dimensional subspace $W \leq \mathbb{F}_2^m$.

For each k -dimensional subspace U containing W , take its full preimage H in P . This group contains P' and projects fully onto each quotient factor, so it projects fully onto each orbit group. Its orbits are exactly the chosen blocks and $|H| = 2^{k+z} = 2^j$. Distinct U 's give distinct subgroups; different partitions give different orbit decompositions. Therefore

$$a_{n,j} \geq \frac{n!}{\delta! 2^a a! (4!)^b b! (8!)^c c!} \left[\begin{matrix} m-4 \\ k-4 \end{matrix} \right]_2. \quad (3)$$

All counted groups inherit the class and exponent restrictions from P .

The Gaussian coefficient satisfies

$$\left[\begin{matrix} R \\ s \end{matrix} \right]_2 = \prod_{i=0}^{s-1} \frac{2^R - 2^i}{2^s - 2^i} \geq 2^{s(R-s)}.$$

Thus the logarithm of the last factor in (3) is at least $(k-4)(m-k) = n^2/16 - O(n)$, uniformly in j . Stirling's formula gives

$$\log_2 \frac{n!}{\delta! 2^a a! (4!)^b b! (8!)^c c!} = (n - b - c) \log_2 n + O(n)$$

uniformly even if b or c is zero: use $b \log b + c \log c = (b+c) \log n + O(n)$, with $0 \log 0 = 0$. Finally,

$$b + c = f/2 + |z - f/2|, \quad n - b - c = 7n/8 - |j - 3n/8| + O(1).$$

Substitution proves (1). The cited published upper bound proves (2).

The upper bound and the eight-point group are prior results. The contribution of this note is the explicit prescribed-order construction and its uniform estimate; whether this refinement is already recorded in the literature has not been established.